Advanced Document Analysis with Large Language Models

Abstract

This study presents a comprehensive pipeline for analyzing long-form documents using state-of-the-art transformer models. Centered on a legal document (Google's Terms of Service), the project covers the stages of text extraction, summarization, segmentation, question generation, and answer generation. By integrating multiple NLP tools, the workflow illustrates scalable and automated document comprehension with real-world applications in compliance, education, and legal informatics.

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1. Introduction

Legal and technical documents are often difficult to parse due to their complexity and length. This project leverages large language models (LLMs) to improve document accessibility through structured summarization and question-answering. The analysis was implemented on the *Google Terms of Service* document and automates the generation of questions and answers per passage to support better understanding and engagement.

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2. Methodology

2.1 Text Extraction

Using the pdfplumber library, the textual content from a multi-page PDF was extracted in full:

```
None
with pdfplumber.open(pdf_path) as pdf:
    for page in pdf.pages:
        extracted_text += page.extract_text()
```

The extracted content was saved to a .txt file (extracted_text.txt) for reproducibility and further NLP tasks.

2.2 Summarization

A pre-trained transformer (t5-small) was employed to produce a concise summary of the first 1000 characters:

```
None
summarizer = pipeline("summarization", model="t5-small")
summary = summarizer(document_text[:1000], max_new_tokens=150,
min_length=30)
```

The summary encapsulated the scope and legal tone of the document effectively.

2.3 Passage Segmentation

To prepare the document for fine-grained analysis, it was split into **26 semantically coherent passages** using nltk's PunktSentenceTokenizer, ensuring each passage stayed under 200 words for token efficiency.

```
None
for sentence in sentences:
    # accumulate passages of ~200 words
```

This ensured compatibility with transformer input limits and maintained contextual relevance within each passage.

2.4 Question Generation

Each of the 26 passages was input to a T5-based question generation model (valhalla/t5-base-qg-hl) in a text2text-generation format.

```
None
qg_pipeline = pipeline("text2text-generation",
model="valhalla/t5-base-qg-hl")
```

```
results = qg_pipeline(f"generate questions: {passage}")
```

Redundancy control and fallback mechanisms were added to ensure a **minimum of 3 quality questions** per passage. The system generated a total of over 75 unique questions.

Example (Passage 1):

- What does the Google Terms of Service cover?
- What are the Terms of Service?
- What should users understand before using Google's services?

2.5 Answer Generation

In a subsequent phase, **answers were generated for the previously generated questions**, utilizing the same or a compatible generative transformer model. Each passage was used as the context to answer its corresponding questions.

```
None
for question in questions:
    answer = qg_pipeline(f"answer question: {question} context:
{passage}")
```

This emulated a closed-book QA setting with contextual grounding, adding a valuable interpretive layer to the pipeline. Each Q&A pair acts as a comprehension unit for the passage.

Example:

- **Q:** What are the Terms of Service?
- **A:** The Terms of Service explain the rules and expectations for using Google services and outline your legal relationship with Google.

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3. Results & Discussion

Passage-Level Output

- **Passages:** 26
- **Total Questions Generated:** 78+ (avg. ~3 per passage)
- **Answer Generation:** 1 answer per question, grounded in passage text

Strengths

- High-quality extraction and coherent summarization of legal content
- Contextual segmentation preserved meaning
- Automated Q&A pairs effectively enhanced document interpretability
- Models ran efficiently on CPU with no GPU acceleration

Observations

- Some questions showed redundancy or mild rephrasing (e.g., “What are the Terms?” vs. “What do the Terms cover?”)
- Answers were generally relevant and well-formed, although some relied heavily on paraphrasing the source text

Limitations

- Input size constraints restricted full-document summarization
- Model answers were not benchmarked for factual accuracy or quality metrics
- No user feedback loop or human-in-the-loop refinement

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4. Conclusion

The experiment demonstrates an effective pipeline for automated document analysis using open-source LLMs. The integration of summarization, segmentation, question generation, and answer generation provides a scalable template for document comprehension tasks. This workflow is well-suited for applications in education, legal services, customer support, and compliance documentation review.

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5. Future Work

- Fine-tune QA models on legal or compliance-specific corpora
- Integrate named entity recognition and topic tagging
- Include evaluation metrics such as BLEU/ROUGE or human scoring
- Deploy as an interactive chatbot or educational aid

References

- Raffel et al., 2020. *Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer*
- Hugging Face Transformers: https://huggingface.co/transformers
- pdfplumber: https://github.com/jsvine/pdfplumber
- NLTK Toolkit: https://www.nltk.org/